

Method for voltage sag source location based on the internal resistance sign in a single-port network

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Abstract: Locating the voltage sag source is to determine on which side of a monitoring device the voltage sag originates. By analysing power network faults, observing from fault points to monitoring devices, power network at both sides of the device can be equivalent to two single-port networks. Based on the polarity of real part of internal impedance in these two single-port networks, the voltage sag source location can be identified accurately. If the real part of internal impedance is positive, the voltage sag source location is consistent with the reference direction, i.e. downstream. If the real part of internal impedance is negative, the voltage sag source location is opposite to the reference direction, i.e. upstream. The calculation of the real part of internal impedance is presented. The method proposed in this study has a solid theoretical basis. Simulation results show the method can locate voltage sag source with an accuracy of 100%, in power networks with various topologies (e.g. radial power system fed from one-source or multiple-source, interconnected power system, ring type network etc.), and the network adopted neutral grounded effectively or ineffectively. It is a practical method with high reliability for voltage sag source location.

1 Introduction

Digital automatic control technologies have been extensively used in industrial production. Compared with conventional equipment, some highly automated equipment is more easily influenced by voltage sag and the requirement for power quality is further more. Whether the system is in normal operation or fault status, it is required to ensure that the fundamental sinusoidal waveform voltage with a small deviation of amplitude can be used, even if several cycles of voltage sags may affect the normal operation of these devices, which will result in tripping or production halt and bring immeasurable economic losses [1, 2]. According to statistics, more than 80% of complaints about power quality problems are voltage sags, while switching overvoltage, harmonics and other complaints account for <20%. At the same time, for power sectors, an increase of complaints and economic disputes caused by voltage sags would weaken their competitiveness in the power market [3]. However, power is a special commodity which is supplied by utilities, but its quality is maintained by both utilities and consumers. For a lack of correct judgement to the reasons of power quality declining, disagreement arises between the power suppliers and users. Only after information about a disturbance location is available, can diagnosis and mitigation strategy be carried out. Moreover, any disputes can be resolved fairly. Therefore, the voltage sag source location has attracted the attention of many scholars in recent years.

Reference [4] is probably the pioneering study about voltage sag source location. On the basis of energy flowing toward the point of failure when a fault is on, the method uses disturbance energy (DE) and the initial peak of the disturbance power (DP) to determine the location of disturbances. The method is simple and easy to implement, but without rigorous theoretical derivation. When DE and DP are small or mismatch, the reliability decreases greatly. Reference [5] is an expansion of this approach which applies instantaneous reactive power to the voltage sag source location by Hilbert transformation. The methods based on the polarity of the slope of system trajectory (SST) [6] and the polarity of the real current component (RCC) [7] are very effective in radial power system fed from either end on symmetrical disturbances, but are

not effective on asymmetrical disturbances, by which the results derived from each phase are always inconsistent so that no conclusion can be drawn from the judgements. Reference [8] proposed a method using the sign of real part of equivalent impedance to locate the voltage sag source, it has a theoretical basis, but it incorrectly assumes the real and imaginary parts of power supply potential $\vec{E}$ are unchanged when calculating the equivalent impedance R_e and results in the wrong calculation of R_e . Therefore, it is not reliable (This will be discussed in the section of simulation verification in this paper.). References [9, 10] define instantaneous voltage and current vector by Clark transformation and apply them to the four methods just mentioned above, which improves the reliability of source location caused by asymmetric disturbance. An approach proposed in [11] uses pre-sag and post-sag impedance magnitude and phase angle measured at monitoring points to locate the voltage sag source. It is not suitable for the radial distribution power system fed from one-source. References [12, 13] raise a method which only needs to measure one quantity to locate voltage sag source in connection with some special occasions, but it is merely used under certain conditions, without universality. After analysing the performance of existing voltage sag source location methods, [14] concludes that: in general, the performance of the methods is much more accurate for symmetrical faults, where accurate results were obtained in 87% of the cases. On the other hand, for asymmetrical faults accurate results were obtained in 65% of the cases. The analysis for meshed grids needs a broader idea of system boundaries. As a result, it is necessary to further research and develop new location method with higher accuracy.

This paper puts forward a new method based on the internal resistance sign (IRS) in a single-port network for voltage sag source location. A theoretical basis of the proposed method will be given in the Section 2. The third chapter shows the method to calculate the real part of internal impedance in a port network. With the help of simulation, partial reasons why the accuracy of existing four methods for voltage sag source location is not high are analysed in Section 4, and the correctness of the new method proposed in this paper is verified. Conclusions of work in this paper will be given in Section 5.

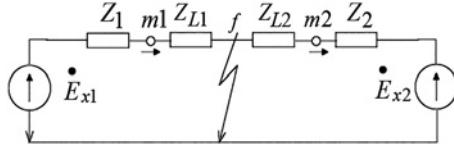


Fig. 1 Short-circuit fault model

2 Theoretical basis

Locating the voltage sag source is to determine on which side of a monitoring device the voltage sag originates. Pre-sag and during-sag three-phase voltage and current waveforms will be captured by power quality monitoring device at monitoring point mi , and then the voltage sag source is located at which side of point mi by analysing the sampled voltage and current.

In power systems, the voltage sag are usually caused by disturbances (e.g. short-circuit fault, large motor starting, capacitor switching etc.). For instance, taking the short-circuit fault as typical disturbance, generally speaking, the probability that two or more short-circuit faults occur simultaneously in power system is usually very low, so only one short-circuit fault is taken into account in this paper and the elements in power system are considered linear. In this case, when fault occurs at point f , the system can always be simplified as shown in Fig. 1.

In Fig. 1, $\dot{E}_{x1}$ and Z_1 are equivalent voltage source and internal impedance of supply-side, $\dot{E}_{x2}$ and Z_2 are equivalent voltage source and internal impedance of customer-side, $x \in [a, b, c]$, a, b, c represents three phases, Z_{L1} and Z_{L2} are line impedances between disturbance source and two monitoring points, respectively, $m1$ and $m2$ are power quality monitoring points on both sides of point f . Observed from the disturbance point f to power quality monitoring points $m1$ or $m2$, the system always can be equivalent to two active single-port networks at $m1$ or $m2$. The reference direction of active current at monitoring point mi can be defined arbitrarily by dotted terminal of the current and voltage transformers installed at monitoring point, which has no relation to the direction of the actual active current, and is generally defined as positive direction when a load consumes active power. Define the internal equivalent impedance of the active single-port network at monitoring point mi as follows

$$Z_{exmi} := \Delta \dot{U}_{xmi} / \Delta \dot{I}_{xmi} = R_{exmi} + jX_{exmi} \quad (1)$$

In formula (1), $\Delta \dot{U}_{xmi} = \dot{U}_{xmi} - \dot{U}_{xpmi}$, $\Delta \dot{I}_{xmi} = \dot{I}_{xmi} - \dot{I}_{xpmi}$, $\dot{U}_{xmi}$ and $\dot{I}_{xmi}$ are voltage and current phasors measured at point mi during the disturbance period, $\dot{U}_{xpmi}$ and $\dot{I}_{xpmi}$ are voltage and current phasors measured at point mi before the disturbance occurs, $x \in [a, b, c]$, a, b, c represents three phases.

According to the positive reference direction of active current at monitoring point mi , if the real part of internal impedance is negative, then the location of disturbance source is consistent with the reference direction of active current, indicating the disturbance source is from downstream; on the contrary, if the real part of internal impedance is positive, the location of the disturbance source is opposite to the reference direction of active current, indicating the disturbance source is from upstream. In summary: (see equation at the bottom of the page)

The theoretical basis of voltage sag source location can be proved by the following inferences.

At monitoring point $m1$, before the fault occurs at point f , in accordance with the arrow ' $\rightarrow$ ' below $m1$ as shown in Fig. 1, the relationship between voltage and current can be described as

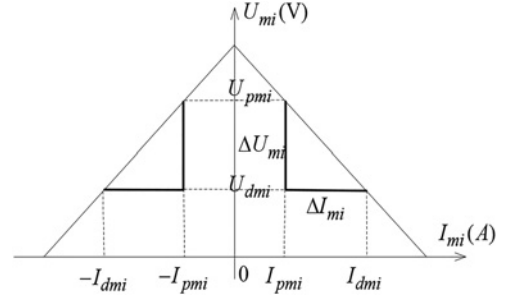


Fig. 2 Equivalent internal impedance characteristic of single-port network at point mi

$$\dot{U}_{xpm1} = \dot{E}_{1x} - \dot{I}_{xpm1} Z_1 \quad (2)$$

where $\dot{U}_{xpm1}$ and $\dot{I}_{xpm1}$ are voltage and current of the x -phase at point $m1$ before the disturbance event. The voltage and current of the x -phase at point $m1$ during the disturbance event can be assumed as $\dot{U}_{xmi}$ and $\dot{I}_{xmi}$, respectively, then

$$\dot{U}_{xmi} = \dot{E}_{1x} - \dot{I}_{xmi} Z_1 \quad (3)$$

In general, the short-circuit capacity of distribution system is not very large so that the power $\dot{E}_{1x}$ in port network can be assumed as ideal power, i.e. $\dot{E}_{1x}$ is constant during the disturbance period. Subtracting (2) from (3), we can find

$$Z_1 = -(\dot{U}_{xmi} - \dot{U}_{xpm1}) / (\dot{I}_{xmi} - \dot{I}_{xpm1}) = -\Delta \dot{U}_{xmi} / \Delta \dot{I}_{xmi} \quad (4)$$

It will be obtained by comparing the definition in formula (1) with formula (4)

$$Z_1 = R_1 + jX_1 = -Z_{exm1} = -R_{exm1} - jX_{exm1} \quad (5)$$

The real part of Z_1 , resistance R_1 is an energy dissipation element, according to the defined reference direction of active current, $R_1 > 0$, there should be $R_{exm1} < 0$. The location of disturbance source is from downstream, consistent with the reference direction of active current. In fact, observed from disturbance source f to point $m1$, on the basis of current reference direction presented in Fig. 1 which flows away from port network, the single-port network has a negative slope of equivalent internal impedance, which can be characterised in the first quadrant of the impedance plane as shown in Fig. 2, so $R_{exm1} < 0$.

At monitoring point $m2$, similarly taking the arrow ' $\rightarrow$ ' below $m2$ presented in Fig. 1 as reference direction, before the fault occurs at point f , the relationship between voltage and current can be described as

$$\dot{U}_{xpm2} = \dot{E}_{2x} + \dot{I}_{xpm2} Z_2 \quad (6)$$

During the disturbance event at point f , the relationship between voltage and current can be described as

$$\dot{U}_{xmi} = \dot{E}_{2x} + \dot{I}_{xmi} Z_2 \quad (7)$$

Subtracting (6) from (7), we can find

$$Z_2 = (\dot{U}_{xmi} - \dot{U}_{xpm2}) / (\dot{I}_{xmi} - \dot{I}_{xpm2}) = \Delta \dot{U}_{xmi} / \Delta \dot{I}_{xmi} \quad (8)$$

It can be obtained by comparing the definition in formula (1) with

$$\begin{cases} R_{exmi} > 0, & \text{opposite to the reference direction, that is, upstream;} \\ R_{exmi} < 0, & \text{consistent with the reference direction, that is downstream.} \end{cases}$$

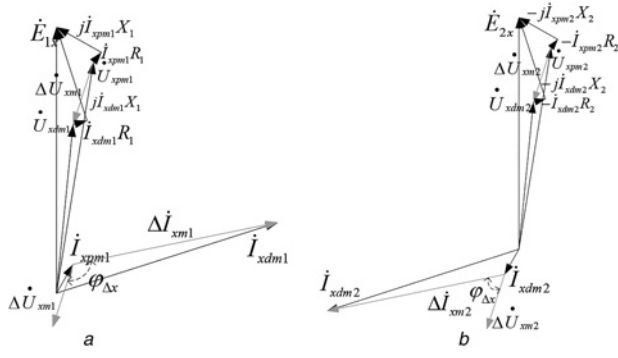


Fig. 3 Phasor diagrams of the single-port networks

a Monitoring point m1
b Monitoring point m2

formula (8)

$$Z_2 = R_2 + jX_2 = Z_{exm2} = R_{exm2} + jX_{exm2} \quad (9)$$

Similarly, the real part of Z_2 , resistance R_2 is also an energy dissipation element, $R_2 > 0$, there should be $R_{exm2} > 0$. The location of disturbance source is from upstream, opposite to the reference direction of active current. In the same way, observed from disturbance source f to point $m2$, on the basis of current reference direction presented in Fig. 1 which flows into port network from outside, the single-port network has a positive slope of equivalent internal impedance, which can be characterised in the second quadrant of the impedance plane as shown in Fig. 2, therefore, $R_{exm2} > 0$.

The judgement method of voltage sag source in this paper can also be verified by the phasor diagrams at points $m1$ and $m2$. In the case that reference direction is defined as shown in Fig. 1, phasor diagram at point $m1$ is presented in Fig. 3a, the angle between the voltage phasor $\Delta \dot{U}_{xm1}$ and current phasor $\Delta \dot{I}_{xm1}$ caused by disturbance satisfies $\varphi_{\Delta xm1} > 90^\circ$ so that $R_{exm1} < 0$ is obtained from the definition of (1), the location of disturbance source f is consistent with the reference direction at point $m1$, that is, downstream. Phasor diagram at point $m2$ is presented in Fig. 3b, the angle between the voltage phasor $\Delta \dot{U}_{xm2}$ and current phasor $\Delta \dot{I}_{xm2}$ caused by disturbance satisfies $\varphi_{\Delta xm2} < 90^\circ$ so that $R_{exm2} > 0$ is obtained from the definition of (1), the location of disturbance source f is opposite to the reference direction at point $m2$, that is, upstream.

Since any meshed structure power network can be modelled as an equivalent network as shown in Fig. 1, the equivalent single-port networks are merely related to the position of disturbance point in the power network, network structure as well as parameters of each element in the power network, and has nothing to do with the actual flow distribution, power sources and loads. Therefore, the new method can be used to any power network structures (such as radial power system fed from one-source or from multiple-source, ring type network etc.), and in neutral grounded effectively or ineffectively as well.

3 Calculation of the real part of internal impedance

Extracting the real part of formula (1) will get

$$R_{exmi} = \Delta U_{xmi} / \Delta I_{xmi} \cos \varphi_{\Delta xmi} \quad (10)$$

where ΔU_{xmi} and ΔI_{xmi} are root mean square (RMS) values of the x -phase disturbance voltage and current; $\varphi_{\Delta xmi}$ is the phase angle difference between the x -phase disturbance voltage and current phasors; $\Delta \dot{U}_{xmi}$ and $\Delta \dot{I}_{xmi}$ are calculated by $\dot{U}_{xpmi}$ and $\dot{I}_{xpmi}$, $\dot{U}_{xpmi}$ and $\dot{I}_{xpmi}$ respectively.

It should be noted that the voltage and current waveforms of pre- and during-events are the measurement value at different

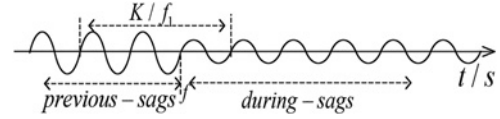


Fig. 4 Diagram of pre-fault and during-fault synchronous sampling

times. To calculate the disturbance voltage and current phasors $\Delta \dot{U}_{xmi}$, $\Delta \dot{I}_{xmi}$, the pre-fault and during-fault voltage and current waveforms must be sampled simultaneously, as shown in Fig. 4. Therefore, the phase-locked loop should be set to start the sampling. In this case, it will be acquired from calculate

$$\begin{cases} \Delta u_{xmi}(n) = u_{xpmi}(n) - u_{xpmi}(n - KN) \\ \Delta i_{xmi}(n) = i_{xpmi}(n) - i_{xpmi}(n - KN) \end{cases} \quad (11)$$

Here, N represents the total number of sampling points in a fundamental cycle T , $n = 0, 1, 2, \dots, K$ is an integer which equals to the difference of fundamental cycles between the during-fault sampling point and the pre-fault sampling point, $K = 1$ or 2 or 3.

Extracting the real and imaginary parts of the disturbance voltage and current phasors $\Delta \dot{U}_{xmi}$ and $\Delta \dot{I}_{xmi}$, respectively, by Fourier analysis, then

$$\begin{cases} \Delta U_{xmi}^{\text{re}} = \frac{1}{N} \sum_{n=0}^{N-1} \Delta u_{xmi}(n) \cdot \sin\left(\frac{2\pi n}{N}\right) \\ \Delta U_{xmi}^{\text{im}} = \frac{1}{N} \sum_{n=0}^{N-1} \Delta u_{xmi}(n) \cdot \cos\left(\frac{2\pi n}{N}\right) \end{cases} \quad (12)$$

$$\begin{cases} \Delta I_{xmi}^{\text{re}} = \frac{1}{N} \sum_{n=0}^{N-1} \Delta i_{xmi}(n) \cdot \sin(2\pi n/N) \\ \Delta I_{xmi}^{\text{im}} = \frac{1}{N} \sum_{n=0}^{N-1} \Delta i_{xmi}(n) \cdot \cos(2\pi n/N) \end{cases} \quad (13)$$

Here, $\Delta U_{xmi}^{\text{re}}$ and $\Delta I_{xmi}^{\text{re}}$ are the real parts of the x -phase disturbance voltage and current phasors $\Delta \dot{U}_{xmi}$ and $\Delta \dot{I}_{xmi}$, $\Delta U_{xmi}^{\text{im}}$ and $\Delta I_{xmi}^{\text{im}}$ are the imaginary parts of the x -phase disturbance voltage and current phasors $\Delta \dot{U}_{xmi}$ and $\Delta \dot{I}_{xmi}$. Put (12) and (13) into (10), R_{exmi} of the x -phase at monitoring point mi can be calculated, $x \in [a, b, c]$

$$R_{exmi} = \frac{(\Delta U_{xmi}^{\text{re}})^2 + (\Delta U_{xmi}^{\text{im}})^2}{\Delta U_{xmi}^{\text{re}} \cdot \Delta I_{xmi}^{\text{re}} + \Delta U_{xmi}^{\text{im}} \cdot \Delta I_{xmi}^{\text{im}}} \quad (14)$$

Parameters of the model shown in Fig. 1 are $Z_1 = 0.2 + j1.8 \Omega$, $Z_2 = 0.25 + j2 \Omega$, $Z_{L1} = 0.3 + j2.6 \Omega$ and $Z_{L2} = 0.2 + j2 \Omega$. The curves of R_{exmi} shown in Fig. 5 is taken Fig. 1 as model, and calculated by (14) with sliding-window algorithm, and the duration of fault is from 0.4 to 0.6 s. $\Delta \dot{U}_{xmi}$ and $\Delta \dot{I}_{xmi}$ are both zero when there is no disturbance, so are the numerators and denominators in formula (14), then we consider $R_{exmi} = 0$. Voltage sag event in the power network is considered to occur if any of three-phase voltage is $< 90\%$ of the rated voltage. Then, R_{exmi} is calculated only for the sag phases, and the judgement for voltage sag source location is made according to the sign of R_{exmi} . In the simulations with different fault types, when a single-phase short-circuit fault occurs, only the fault-phase voltage is $< 90\%$ of the rated voltage, so we merely calculate R_{exmi} of the fault phase, as shown in Figs. 5a and b; when two-phase ground fault and two-phase short-circuit fault occur, the voltage of these two phases will be $< 90\%$ of rated voltage, respectively, so R_{exmi} of these two fault phases will be calculated and the results are shown in Figs. 5c-f. Since the calculation results of R_{exmi} are the same, only one-phase R_{exmi} is plotted; three-phase voltages are all $< 90\%$ of rated voltage when the three-phase short-circuit fault occurs, so R_{exmi} of three phases will be calculated. Only one phase R_{exmi} is plotted since the three-phase results of R_{exmi} are equal, just as shown in Figs. 5g

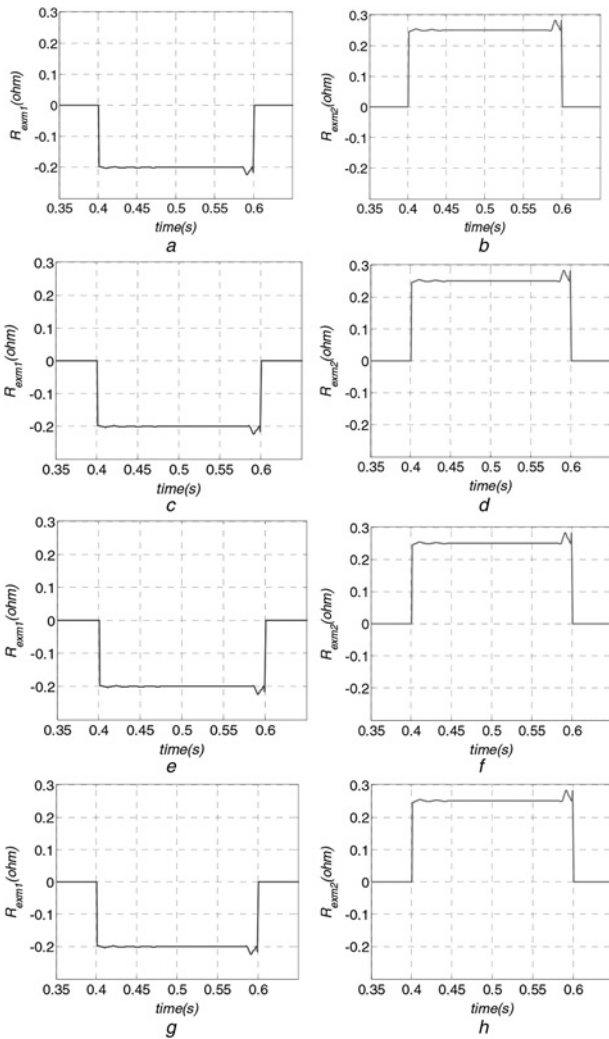


Fig. 5 Real part of internal impedance in equivalent port network with various faults

- a Single-phase short-circuit fault
- b Single-phase short-circuit fault
- c Two-phase ground fault
- d Two-phase ground fault
- e Two-phase short-circuit fault
- f Two-phase short-circuit fault
- g Three-phase short-circuit fault
- h Three-phase short-circuit fault

and h . It can be seen from these graphs that the sign of R_{exmi} , the real part of internal impedance can accurately determine the location of voltage sag source. Figs. 5a, c, e and g show the curves of the real part of internal impedance measured at monitoring point $m1$ when four types of fault occur. R_{exmi} are all negative, indicating that the disturbance source is from downstream (consistent with the reference direction of point $m1$); Figs. 5b, d, f and h show the curves of the real part of internal impedance measured at monitoring point $m2$ when four types of fault occur, R_{exmi} is positive, indicating that the disturbance source is from upstream (opposite to the reference direction of point $m2$); it can also be seen from the R_{exmi} curves in Fig. 5 that steady-state values of R_{exmi} are equal under these four fault conditions, just as R_{exm1} calculated from point $m1$ (Figs. 5a, c, e and g) is -0.2Ω , i.e. $R_{exm1} = -R_1$; while R_{exm2} calculated from point $m2$ (Figs. 5b, d, f and h) is 0.25Ω , i.e. $R_{exm2} = R_2$; these curves verify the correctness of the theory proposed in this paper directly: namely, the magnitude of R_{exmi} is equal to the internal resistance in the equivalent port network at point mi , which is observed from voltage sag source to the monitoring point mi and the sign of R_{exmi} is decided by the reference direction of point mi , which will be

negative when reference direction flows away from port network while positive when reference direction flows into port network from outside.

In summary, the proposed method for voltage sag source location can be implemented with the procedure as follows:

(i) Measure the voltage and current of each phase in normal operation, and calculate the RMS values of the phase voltage so as to determine whether there is a voltage sag event or not. With respect to the power network in which neutral is grounded ineffectively, when single-phase ground fault occurs, three-phase line voltage is still balanced and has no influence on the operation of electrical equipment. To improve the reliability of power supply, there is no need to isolate the fault point immediately and the power system can still operate normally. Therefore, this situation is not regarded as there is a voltage sag event which affects the operation of electrical equipment. As a result, the judgments of determining whether voltage sag event occurs or not in this paper will be divided into two cases.

For neutral grounded effectively power network, the phase-to-ground voltages will be measured and calculated by (15)

$$U_{xmi} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} [u_{xmi}(n)]^2} \quad (15)$$

where U_{xmi} is the RMS values the x phase-to-ground voltage measured at monitoring point mi . If any one of the three phase-to-ground voltages U_{xmi} is $<90\%$ of rated voltage represents voltage sag event occurs.

For neutral grounded ineffectively power network, the phase-to-neutral voltages will be measured and calculated by (16)

$$U_{xmi} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} [u_{xmi}(n) - u_{mi}^0(n)]^2} \quad (16)$$

In (16), $u_{mi}^0(n) = [u_{ami}(n) + u_{bmi}(n) + u_{cmi}(n)]/3$ represents the zero-sequence voltage in power network. If any one of three phase-to-neutral voltages U_{xmi} is $<90\%$ of rated phase voltage, the voltage sag event occurs.

- ii. Record the starting time of voltage sag, and measure the voltage and current of each phase continuously when voltage sag occurs in the power network.
- iii. Find out which phase voltage is $<90\%$ of rated phase voltage and calculate the real part of internal impedance – of that phase measured at monitoring point mi by (14).
- iv. Locate voltage sag source according to the sign of R_{exmi} , the source is opposite to the reference direction if R_{exmi} is positive, indicating that source is in the upstream of monitoring point mi ; on the contrary, the source is consistent with the reference direction if R_{exmi} is negative, indicating that source is in the downstream of monitoring point mi .

4 Simulation verification and analysis

4.1 Radial system

4.1.1 Parameters of the test network model and simulation results: Fig. 6 shows a radial system, E1 is an infinite power supply whose short-circuit capacity is 1000 MVA. E2 is selected as the type of QF-60-2 which generates 54 MW active power actually, $x_d' = 0.21$. Capacities of three transformers are 6.3, 1.25 and 1 MVA, respectively, their coupling modes and connected loads are just shown in Fig. 6. Among them, a load connected to transformer T3 is controlled rectifier power supply and is a non-linear load. The 110 kV side and 380 V side are both neutral grounded power networks and 10 kV side is neutral non-grounded

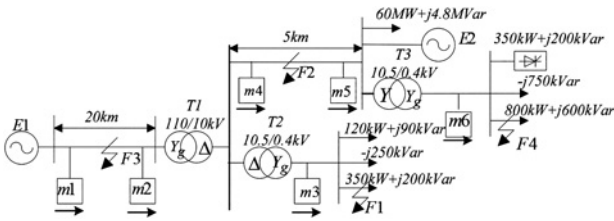


Fig. 6 Radial system

power network. Four fault points are set at F1, F2, F3 and F4 in the simulation model. Four fault types are simulated at each point including three-phase short-circuit fault, two-phase short-circuit fault, two-phase ground fault and single-phase ground fault. Fault types are presented in the ‘()’ in Table 1 such as: F1(3), F1(2), F3(1,1) and F4(1) represents three-phase short-circuit fault at point F1, two-phase short-circuit fault at point F1, two-phase ground fault at point F3 and single-phase ground fault at point F4, respectively. There are six monitoring points $m1$, $m2$, $m3$, $m4$, $m5$ and $m6$ and their reference directions are ‘→’ below the monitoring devices mi just as shown in Fig. 6.

Five methods for voltage sag source location have been tested in the same network model as shown in Fig. 6. These methods are DE and power (DEP) method [4], the SST method [6], the RCC method [7], the RS of the equivalent impedance method [8] and the proposed IRS in this paper. The simulation results obtained from all methods are shown in Table 1. Symbol ‘*’ in Table 1 indicates that the amplitudes of three-phase voltages are >90% of rated phase voltage: namely, voltage sag event did not happen at monitoring point mi according to the definition of voltage sag so that judgement for the voltage sag source location cannot be made. Voltage sag event occurs in power network if any one of three-phase voltages observed at monitoring point mi is <90% of the nominal voltage, then judgement for voltage sag source location will be made; ‘-’ indicates that the results of three phases are not in agreement and no result can be got: namely, failed judgements; ‘+’ indicates that the judgements for voltage sag source of fault phase are consistent and correct, if the phase, in which voltage is <90% of rated voltage, is just judged by the method proposed in original document, despite the judgement results of three phases are inconsistent; ‘↓’ means the downstream, ‘↑’ means the upstream, red arrow represents incorrect judgements. Each method is simulated 96 times including 33 times simulation cases where at least one phase voltage measured at monitoring points is <90% of rated phase voltage and then judgement for voltage sag source is given.

4.1.2 Analyses and comparison with the exiting methods: Simulation results show that: for the downstream voltage sag disturbance in radial power system fed from

one-source, what is DEP method [4] measured is the difference between the pre-sag and during-sag energy and power delivered at monitoring points mi and is essentially a difference between the active power loss on line resistance from mi to the fault (short circuit) point and active power delivered at monitoring point mi . Therefore, if monitoring point mi is relatively close to fault (short circuit) point: namely, line resistance is small, there would be a small corresponding active power loss or if active power delivered in normal operation at point mi is relatively large, DP and energy would be negative which makes an error judgement. That is why the DEP method has quite a high error rate for downstream voltage sag disturbance in radial power system fed from one-source. As a result of downstream disturbances of F1, F2 and F4 monitored at points $m3$, $m4$ and $m6$, respectively, as shown in Fig. 6, the accurate result is zero.

The RCC method [7] and the SST method [6] both have a high accuracy for the identification of symmetrical three-phase voltage sag source that are 97 and 100%, respectively, which agrees basically with the simulation analysis conclusion 87% given in [14]. For voltage sags caused by asymmetric faults, the RCC method RCC [7] and the SST method [6] neither give simulation and experimental tests, nor present detailed implemental steps and criterion in their original documents. As mentioned above, sag event occurs in the power system if any of three-phase voltages is <90% the nominal voltage. The methods and steps which are also according to the original literature [6, 7] are taken to simulate. The simulation results are: the probability that three-phase judgements are inconsistent, drawing no conclusions are 62.5 and 45.8%; wrong judgements are 0 and 8.3%, and correct judgements are 37.5 and 45.8%. According to the definition of voltage sag, the phase, in which voltage is <90% of rated voltage, is just judged by the method proposed in original document. If the results of two-phase or one-phase are consistent, then judgements can be obtained; otherwise, it is still considered to draw no conclusion, the results are as follows: no conclusion are 45.8 and 16.7%; wrong judgements are 0 and 8.3%; and correct judgements are 54.2 and 75%.

An RS-based method is proposed in [8]. When calculating R_e using this method, the real and imaginary parts of power supply potential $\vec{E}_x$ are both assumed unchanged. In fact, the power supply potential E_x is assumed as a constant, whereas $\vec{E}_x$ is a phasor with a phase angle, as a result the fundamental positive-sequence values of the real and imaginary parts decomposed by Fourier analysis are not equal during different

Table 2 Location accuracy of various methods

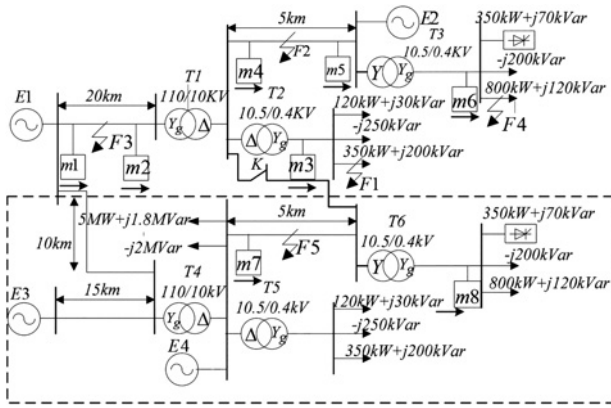
Location method	DEP	SST	RCC	RS	IRS
Accuracy, %	66.7	81.8	63.7	75.8	100

Table 1 Summarised simulation results of five methods for sag source location: DEP/SST/RCC/RS/IRS

Fault types	DEP/SST/RCC/RS/IRS					
	$m1$	$m2$	$m3$	$m4$	$m5$	$m6$
F1(1)	*/**/**	*/**/**	↑/+/-/↓	*/**/**	*/**/**	*/**/**
F1(1,1)	*/**/**	*/**/**	↑/+/-/↓	↑/↑/-/↑	*/**/**	*/**/**
F1(2)	*/**/**	*/**/**	↑/+/-/↓	↑/↑/↑/↑	*/**/**	*/**/**
F1(3)	*/**/**	*/**/**	↑/↓/-/↓	↑/↑/↑/↑	*/**/**	*/**/**
F2(1)	*/**/**	*/**/**	*/**/**	*/**/**	*/**/**	*/**/**
F2(1,1)	*/**/**	*/**/**	↑/↑/↑/↑	↑/↓/-/↓	*/**/**	*/**/**
F2(2)	*/**/**	*/**/**	↑/↑/↑/↑	↑/↓/-/↓	*/**/**	*/**/**
F2(3)	*/**/**	*/**/**	↑/↑/↑/↑	↑/↓/↑/↓	*/**/**	*/**/**
F3(1)	↓/↓/+/-/↓	↑/↓/+/-/↑	↑/↑/↑/↑	↑/-/-/↑	*/**/**	*/**/**
F3(1,1)	↓/↓/-/↓/↓	↑/↓/+/-/↑	↑/↑/↑/↑	↑/-/-/↑	*/**/**	*/**/**
F3(2)	↓/+/-/↓/↓	↑/-/↑/↑	↑/↑/↑/↑	↑/-/-/↑	*/**/**	*/**/**
F3(3)	↓/↓/↑/↑	↑/↑/↑/↑	↑/↑/↑/↑	↑/↑/↑/↑	*/**/**	*/**/**
F4(1)	*/**/**	*/**/**	*/**/**	*/**/**	*/**/**	↑/+/-/↑
F4(1,1)	*/**/**	*/**/**	*/**/**	*/**/**	*/**/**	↑/+/-/↑
F4(2)	*/**/**	*/**/**	*/**/**	*/**/**	*/**/**	↑/+/-/↑
F4(3)	*/**/**	*/**/**	*/**/**	*/**/**	*/**/**	↑/↓/↓/↓

Table 3 Simulation results of IRS

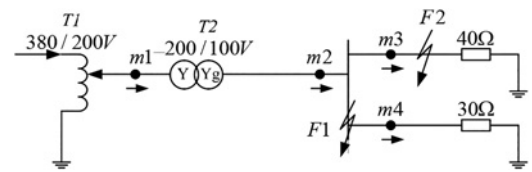
Fault types	F(1)/F(1,1)/F(2)/F(3)							
	m1	m2	m3	m4	m5	m6	m7	m8
F1	*/**/*	*/**/*	↓/↓/↓	*/**/*	*/**/*	*/**/*	*/**/*	*/**/*
F2	*/**/*	*/**/*	*/1/1/1	*/↓/↓/↓	*/1/1/1	*/1/1/1	*/**/*	*/1/1/1
F3	↓/↓/↓	↑/1/1/1	↑/1/1/1	↑/1/1/1	*/**/*	*/**/*	*/1/1/1	↑/1/1/1
F4	*/**/*	*/**/*	*/**/*	*/**/*	*/**/*	↓/↓/↓	*/**/*	*/**/*
F5	*/**/*	*/**/*	*/1/1/1	*/1/1/1	*/**/*	*/**/*	*/↓/↓/↓	*/1/1/1

**Fig. 7** Non-radial system

periods. They should follow the transformation relationship in formula (17)

$$\begin{bmatrix} E_x^{\text{re}} \\ E_x^{\text{im}} \end{bmatrix} = \begin{bmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{bmatrix} \begin{bmatrix} E_x^{\text{re}} \\ E_x^{\text{im}} \end{bmatrix}, \quad x \in \{a, b, c\} \quad (17)$$

where E_x^{re} and E_x^{im} are the real and imaginary parts of power supply potential $\dot{E}_x$ whose initial phase angle is φ , E_x^{re} and E_x^{im} are the real and imaginary parts of power supply potential $\dot{E}_x$ whose initial phase angle is φ' , and $\beta = \varphi' - \varphi$ is satisfied. Since the real and imaginary parts of power source potential $\dot{E}_x$ decomposed according to the literature [8] are changeable, thus, the formulas (13) and (14) in [8] is always an 'underdetermined' equation so that R_e has not determined 'the unique solution'. It can be seen that the judgement based on the polarity of R_e is not reliable. Actually, the authors in literature [8] has already found: 'the impedance-based method is theoretically sound, but there is one major difficulty in its practical implementation: the results could be inconsistent if different pre- or during-event cycles are selected'.

**Fig. 9** Laboratory testing circuit

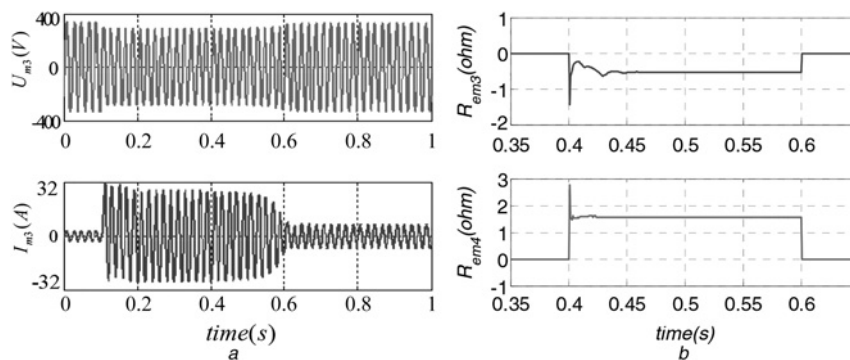
The accuracy rate of the five location methods and simulation results of IRS are shown in Tables 2 and 3. Simulation results show that: the method proposed in this paper can locate voltage sag source deterministically: namely, the correct rate is 100%, indicating a very high reliability.

4.2 Non-radial system

Power supply model as shown in Fig. 7 is utilised to verify that the method proposed in this paper is also applicable to non-radial system (such as radial power system from multiple-source, ring type network etc.). Simulation parameters of elements involved in the dashed box in Fig. 7 are set as that in Fig. 6. The power supply model provided in Fig. 7 is used to simulate on the proposed method and the location accuracy is 100% (some other available methods are not applicable to the ring type network; thus, the judgement and simulation results are not given any more). It is proved that the proposed method is applicable to any network structure.

4.3 Motor starting

In Fig. 6, replace the 350 + j200 kVA load connected to transformer T2 with an inductive motor which has an equal capacity, and in this way, a large motor starting simulation model is constituted. The voltage and current waveforms in Fig. 8a are obtained from observation point $m3$ when an inductive motor is put into operation at 0.4 s. Fig. 8b reveals the real part value of the internal impedance measured at monitoring point $m3$ and $m4$ during motor starting. R_{em3} is negative, indicates that the disturbance source is

**Fig. 8** Inductive motor starting

a Voltage and current waveforms at $m3$

b Internal resistance in the port network at $m3/m4$

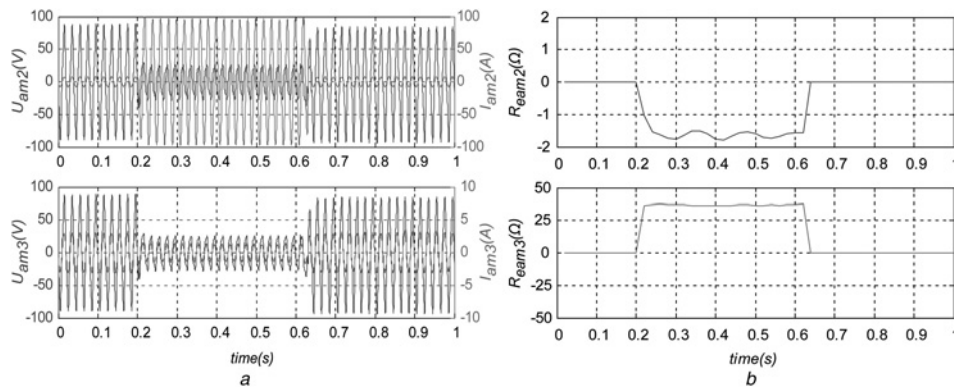


Fig. 10 Sampled *a*-phase voltage and current waveforms at points *m2* and *m3*, respectively, when a two-phase (phases *a* and *c*) short-circuit fault occurs at point *F1*

a Sampled voltage and current waveforms
b Real parts of internal impedances

Table 4 Laboratory testing results of IRS

Fault types	F(1)/ F(1,1)/ F(2)/ F(3)			
	<i>m1</i>	<i>m2</i>	<i>m3</i>	<i>m4</i>
F1	↓/↓/↓/↓	↓/↓/↓/↓	↑/↑/↑/↑	↑/↑/↑/↑
F2	↓/↓/↓/↓	↓/↓/↓/↓	↓/↓/↓/↓	↑/↑/↑/↑

Table 5 Laboratory testing results of IRS

Fault types	F(1)/F(1,1)/F(2)/F(3)					
	<i>m1</i>	<i>m2</i>	<i>m3</i>	<i>m4</i>	<i>m5</i>	<i>m6</i>
F1	↓/↓/↓/↓	↓/↓/↓/↓	↑/↑/↑/↑	↑/↑/↑/↑	↓/↓/↓/↓	↑/↑/↑/↑
F2	↓/↓/↓/↓	↓/↓/↓/↓	↓/↓/↓/↓	↑/↑/↑/↑	↓/↓/↓/↓	↑/↑/↑/↑
F3	↓/↓/↓/↓	↓/↓/↓/↓	↑/↑/↑/↑	↑/↑/↑/↑	↓/↓/↓/↓	↑/↑/↑/↑

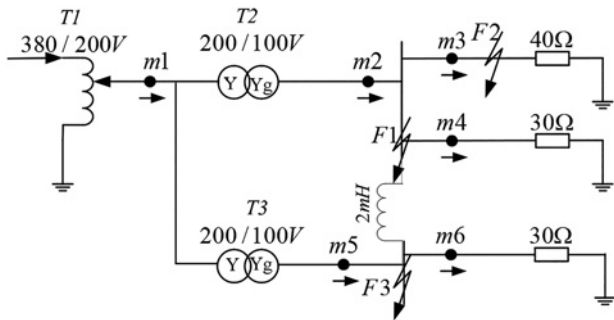


Fig. 11 Laboratory testing circuit

located in the downstream of *m3*, R_{em4} is positive, indicates that the disturbance source is located in the upstream of *m4*. This result is in agreement with the practical situation. Voltage sags measured at other observation points are quite small, which dissatisfy the definition range of voltage sag source (<90% of rated phase voltage), so there is no judgement about them.

4.4 Laboratory test verification

4.4.1 Radial network: Fig. 9 represents the laboratory testing circuit. In Fig. 9, the automatic coupling transformer T1 reduces power supply voltage from 380 to 200 V and the isolation transformer T2 transfers the voltage from 200 to 100 V. Define the active power consumption as positive reference direction at each monitoring point (as '→' shown in Fig. 9). Fig. 10*a* is the sampled *a*-phase voltage and current waveforms at points *m2* and *m3*, respectively, when a two-phase (phases *a* and *c*) short-circuit fault occurs at point *F1*. Fig. 10*b* is the real parts of internal impedances measured at point *m2* and *m3*. The results $R_{m2} < 0$ and $R_{m3} > 0$ indicate that the disturbance source locates in the downstream of *m2* and the upstream of *m3*. This conclusion is consistent with the actual situation. Testing results are presented in Table 4 and the location accuracy is 100%.

4.4.2 Ring type network: Fig. 11 represents the hand-in-hand testing circuit in laboratory. Fig. 12*a* is the sampled voltage and current waveforms at *m2* and *m3*, respectively, when a two-phase short-circuit fault occurs at point *F1*. Fig. 12*b* is the real parts of

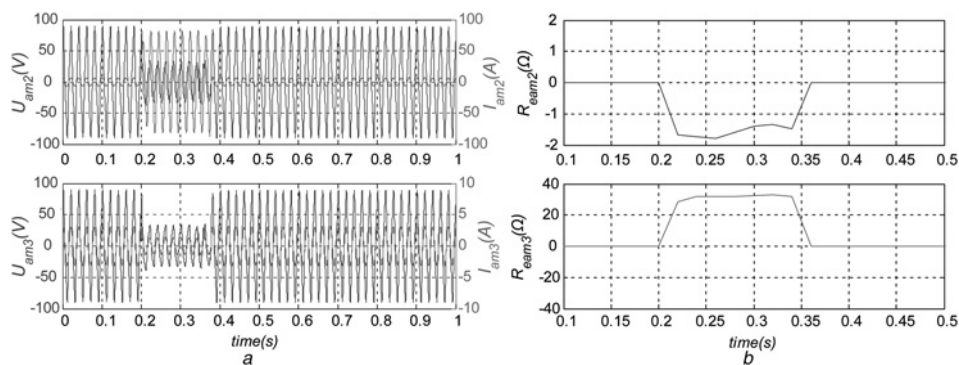


Fig. 12 Sampled voltage and current waveforms at *m2* and *m3*, respectively, when a two-phase short-circuit fault occurs at point *F1*

a Sampled voltage and current waveforms
b Real parts of internal impedances

internal impedances measured at $m2$ and $m3$. The results $R_{m2} < 0$ and $R_{m3} > 0$ indicate that the disturbance source is in the downstream of $m2$ and in the upstream of $m3$, respectively, which is consistent with the actual situation. Testing results are presented in Table 5 and the location accuracy is 100%.

5 Conclusions

On the basis of the linear active single-port theory, this paper presents a method using the polarity of the real part of internal impedance in single-port network to locate the voltage sag source. The main conclusions of this paper can be summarised as follows:

(i) By measuring the voltage and current waveforms at monitoring point and sampling them synchronously according to the fundamental frequency of the power system, the disturbance voltage and current can be obtained and the real part of internal impedance in the single-port network is calculated. If the real part of internal impedance is positive, the location of voltage sag source is consistent with the reference direction, i.e. downstream; if the real part of internal impedance is negative, the location of voltage sag source is opposite to the reference direction, i.e. upstream. Location method proposed in this paper must sample voltage and current waveforms synchronously according to the fundamental frequency of the power system when calculating the disturbance voltage and current.

(ii) The proposed voltage sag source location method here is based on the theory of linear single-port network and has a solid theoretical basis, as a result, it can locate the voltage sag source deterministically, and namely the correct rate is 100%. Since this location method merely utilises the fundamental voltage and current to calculate and make judgement, besides it can be seen from Fig. 1 and the definition in formula (1) that the calculation value of the equivalent internal impedance in port network has nothing to do with load, as long as the disturbance of the fundamental voltage and current at monitoring point can be measured accurately, then the real part of the internal impedance will be calculated accurately. As such, the proposed method is also applicable in circuits with non-linear loads.

(iii) Since any meshed structure can be modelled as an equivalent single-port network as shown in Fig. 1, equivalent single-port network is merely related to the position of disturbance point in the power system, network structure as well as parameters of each element in the network. The network is independent of the actual power distribution and direction of the actual active current; therefore, this presented method is effective to be used not only to any network structures (such as radial power system,

interconnected power system, ring type network etc.), but also to the meshed power network with neutral grounded effectively or ineffectively.

(iv) The reference direction used for locating voltage sag source just depends on the dotted terminal of the voltage and current transformers installed at monitoring point, and has no relation to the direction of the actual active current in the network. Active power consumption of load is generally defined as positive; otherwise it is negative.

(v) Though the conclusion is deduced by the analysis of fault disturbances occurring in a power system, this method is also applicable to locate voltage sag source caused by other disturbances (such as: capacitor switching, transformer switching, large motor starting etc.).

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